MODERN PHYSICS – MIDTERM 1.

Please write every problem on separate sheet of paper, and write your Neptun code and Name on every paper you submit. Make sure to write all the formulas that you evaluate, as well as your calculations, so you can get partial score even if there is an error. Give the answers with a precision of 3 significant digits, together with the units, unless otherwise specified. Note that if a problem is scored, e.g., X + Y + Z points, then *three* answers are expected. Read and start every part of the problems, you can still get partial score even if you don't have the earlier results.

Physical constants

$$m_e = 9.11 \times 10^{-31}\,\mathrm{kg} \qquad h = 6.626 \times 10^{-34}\,\mathrm{J\,s} = 4.136 \times 10^{-15}\,\mathrm{eV\,s} \qquad c = 3.00 \times 10^{8}\,\mathrm{m/s}$$

$$m_e c^2 = 511\,\mathrm{keV} \qquad hc = 1.986 \times 10^{-25}\,\mathrm{J\,m} = 1240\,\mathrm{eV\,nm}$$

$$E_1 = \frac{h^2}{8 m_e L^2} \text{ (ground state energy for a free electron in a box)}$$

1. Photoelectric effect (2 + 5 + 3 = 10 points)

A Thungsten (W) cathode is irradiated with UV light of wavelength $\lambda = 206.5\,\mathrm{nm}$. As the light intensity is changed, the instrument consistently measures a stopping voltage $-V_0 = 1.5\,\mathrm{V}$.

(a) What is the energy of the E_γ of the incident photons? (2pt)

(b) What is the *maximum* kinetic energy K_{max} (2pt) and the corresponding *minimum* de Broglie wavelength λ_{dB} (3pt) of the photoemitted electron?

(c) What is the work function of the W cathode ϕ_{W}? (3pt)

Assume the electron can be described non-relativistically.

2. Absorption and emission processes (4 + 3 + 3 = 10 points)

An electron is trapped in an infinite square-well potential of width $L = 1.0\,\mathrm{nm}$. The electron is initially in the ground state ($n = 1$) and is excited to a higher-energy state ($n = 4$) by absorbing a photon. After some time, the electron decays to the state n_f emitting a photon with energy $E_\gamma = 4.5\,\mathrm{eV}$.

(a) Calculate the energy of the states E_1 and E_4? (2+2pt)

(b) What is the wavelength λ of the *absorbed* photon in the process $1 \to 4$? (3pt)

(c) What is the final state n_f of the electron? (3pt)

3. Quantum-mechanical expectation values (2 + 2 + 10 + 3 + 2 = 19 points)

Consider the following wave function for a free electron in a box (infinite potential well), defined in the interval $[-1, 1]$ measured in appropriate SI units.
Give the exact results in terms of $\hbar$ and m_e, no need to write the units in the results.

$$\psi(x) = A(x^3 + ax)$$

(a) What are the appropriate boundary conditions? Find the value of a that fulfills the boundary conditions. (1+1pt)

(b) Show that (for the above value of a) the normalization condition is fulfilled for $A = \sqrt{105}/4$. (2pt)

(c) Calculate the expectation values of $\langle x \rangle$, $\langle x^2 \rangle$, $\langle p \rangle$, and $\langle p^2 \rangle$ in the normalized state found above. (2+3+2+3pt)
Hint: It is easy to make a mistake while simplifying fractions, you might want to double-check your results using a calculator. All the results simplify to fractions with single-digit denominators.

(d) Compute Δx, Δp and check that the Heisenberg uncertainty relation is satisfied. (3pt)

(e) Using $\langle p^2 \rangle$ compute $\langle E \rangle$ for this state. Compare the value to the energy E_2 of the first excited eigenstate of the potential well. (2pt)

4. The $2s$ state of Hydrogen (3 + 5 + 3 = 11 points)
The radial part of the wavefunction of the $2s$ state of the Hydrogen atom (with quantum numbers $n = 2$ and $\ell = 0$) reads

$$R_{20}(r) = \frac{1}{2\sqrt{2}a_0^{3/2}}\left(2 - \frac{r}{a_0}\right)e^{-\frac{r}{2a_0}}.$$

Unlike the ground state wavefunction, this state displays a *forbidden* radius (node) r_0 for which the probability density to find the electron at a distance r_0 from the nucleus is zero.

(a) Find the position of the node r_0. (3pt)

(b) Find the most probable value $\bar{r}$ that maximizes the radial probability density. (5pt)
Hint: Maximizing the natural logarithm of the radial probability density is simpler than brute-force maximizing the function itself. Define $x = r/a_0$, and use

$$\frac{d}{dx}\log\left(f(x)\right) = \frac{f'(x)}{f(x)} = 0 \qquad \Longrightarrow \qquad f'(x) = \frac{d}{dx}f(x) = 0.$$

This procedure yields solutions $r_\pm$ (two local maxima, one of which is the global maximum) while the solutions ($r = 0$, and the node $r = r_0$) corresponding to minima are excluded.

(c) Sketch the radial probability density function, aiming at getting the overall shape and the relative positions of the node and the maxima correctly. (3pt)

Give the answers for r_0 and $\bar{r}$ parametrically in terms of the Bohr radius a_0.

Hint: The volume integral in spherical coordinates is given by:

$$\int dV = \int_0^\infty \mathrm{d}r\, r^2 \int_0^\pi \mathrm{d}\theta \sin(\theta) \int_0^{2\pi} \mathrm{d}\phi.$$

OFFICIAL SOLUTIONS

1. Photoelectric effect (2 + 5 + 3 = 10 points)
A Thungsten (W) cathode is irradiated with UV light of wavelength $\lambda = 206.5\,\mathrm{nm}$. As the light intensity is changed, the instrument consistently measures a stopping voltage $-V_0 = 1.5\,\mathrm{V}$.

(a) What is the energy of the E_γ of the incident photons? (2pt)

(b) What is the *maximum* kinetic energy $K_{\max}$ (2pt) and the corresponding *minimum* de Broglie wavelength λ_{dB} (3pt) of the photoemitted electron?

(c) What is the work function of the W cathode ϕ_{W}? (3pt)

Assume the electron can be described non-relativistically.

Solution
The energy of the incident photon is given by

$$E_\gamma = pc = \frac{hc}{\lambda} \approx 6.00\,\mathrm{eV}.$$

The maximum kinetic energy is related to the stopping voltage

$$K_{\max} = -eV_0 = 1.5\,\mathrm{eV}$$

and the corresponding minimum de Broglie wavelength is given by

$$\lambda_{\mathrm{dB}} = \frac{h}{p} = \frac{h}{\sqrt{2m_e K}} = \frac{hc}{\sqrt{2m_e c^2 K}} \approx 1.00\,\mathrm{nm}.$$

The work function of the cathode is found imposing energy conservation between the incoming photon and the photoemitted electron

$$E_\gamma = \phi_{\mathrm{W}} + K_{\max} = \phi_{\mathrm{W}} - eV_0,$$

yielding $\phi_{\mathrm{W}} \approx 4.5\,\mathrm{eV}$.

2. Absorption and emission processes (4 + 3 + 3 = 10 points)
An electron is trapped in an infinite square-well potential of width $L = 1.0\,\mathrm{nm}$. The electron is initially in the ground state ($n = 1$) and is excited to a higher-energy state ($n = 4$) by absorbing a photon. After some time, the electron decays to the state n_f emitting a photon with energy $E_\gamma = 4.5\,\mathrm{eV}$.

(a) Calculate the energy of the states E_1 and E_4? (2+2pt)

(b) What is the wavelength λ of the *absorbed* photon in the process $1 \to 4$? (3pt)

(c) What is the final state n_f of the electron? (3pt)

Solution
The energy levels of the infinite square-well potential problem are given by

$$E_n = \frac{h^2}{8m_e L^2}\, n^2 = \frac{(hc)^2}{8m_e c^2 L^2}\, n^2,$$

yielding $E_1 = 0.375\,\mathrm{eV}$ and $E_4 = 6.0\,\mathrm{eV}$, obtained using $hc = 1240\,\mathrm{eV\,nm}$ and $m_e c^2 = 511\,\mathrm{keV}$.
The wavelength is obtained by considering that the energy of the photon is

$$E_\gamma^{1\to 4} = E_4 - E_1 = 5.625\,\mathrm{eV}$$

and that, according to de Broglie

$$\lambda = \frac{h}{p} = \frac{hc}{pc} = \frac{hc}{E_\gamma^{1\to 4}} \simeq 220\,\mathrm{nm}.$$

In the photon emission

$$E_\gamma^{4\to n_f} = E_4 - E_{n_f} = \frac{(hc)^2}{8m_e c^2 L^2}(16 - n_f^2) = 4.5\,\mathrm{eV}$$

yielding $n_f = 2$.

3. Quantum-mechanical expectation values (2 + 2 + 10 + 3 + 2 = 19 points)
Consider the following wave function for a free electron in a box (infinite potential well), defined in the interval $[-1, 1]$ (measured in appropriate SI units, no need to write the units in the results).

$$\psi(x) = A(x^3 + ax)$$

(a) What are the appropriate boundary conditions? Find the value of a that fulfills the boundary conditions. (1+1pt)

(b) Show that (for the above value of a) the normalization condition is fulfilled for $A = \sqrt{105}/4$. (2pt)

(c) Calculate the expectation values of $\langle x\rangle$, $\langle x^2\rangle$, $\langle p\rangle$, and $\langle p^2\rangle$. (2+3+2+3pt)

(d) Compute Δx, Δp and check that the Heisenberg uncertainty relation is satisfied. (3pt)

(e) Using $\langle p^2\rangle$ compute $\langle E\rangle$ for this state. Compare the value to the energy E_2 of the first excited eigenstate of the potential well. (2pt)

Give the exact results in terms of $\hbar$ and m_e.

Solution
(a) The boundary conditions for this problem are $\psi(-1) = \psi(1) = 0$, both yielding $a = -1$.

(b) For $a = -1$, the wave function reads

$$\psi(x) = A(x^3 - x).$$

The normalization condition reads

$$1 = |A|^2 \int_{-1}^{1} \mathrm{d}x\,(x^3 - x)^2 = |A|^2 \cdot 2 \int_0^1 \mathrm{d}x\,(x^6 - 2x^4 + x^2) = |A|^2 \cdot 2\left[\frac{1}{7}x^7 - \frac{2}{5}x^5 + \frac{1}{3}x^3\right|_0^1 = |A|^2 \frac{16}{105}$$

which yields $A = \sqrt{105}/4$, as requested to show.

(c) The expectation value of the position operator is

$$\langle x\rangle = \frac{105}{16}\int_{-1}^{1} \mathrm{d}x(x^3 - x)x(x^3 - x) = 0,$$

and the expectation value of the momentum operator is performed using the relation $p = -i\hbar\partial/\partial x$, yielding

$$\langle p\rangle = -i\hbar\frac{105}{16}\int_{-1}^{1} \mathrm{d}x(x^3 - x)\,\frac{\partial}{\partial x}(x^3 - x) = 0,$$

Both are zero by symmetry, because they are integrals of an odd function on a symmetric domain. The same conclusion can be obtained by performing the integrals explicitly.

Clarification: the momentum operator on a finite domain with Dirichlet boundary conditions, i.e., $\psi(x) = 0$ at the boundaries (such as in this case) is formally Hermitian and satisfies the condition $\langle\phi|p\psi\rangle = \langle p\phi|\psi\rangle$. Therefore, its expectation value must be real, and the integral above must vanish. However, note that the operator $p = i\hbar\partial/\partial x$ is not self-adjoint (i.e., $p \neq p^\dagger$), and under some conditions, it can have undefined or formally complex expectation values.

The expectation value of x^2 is

$$\langle x^2\rangle = \frac{105}{16}\int_{-1}^{1} \mathrm{d}x(x^3 - x)x^2(x^3 - x) = \frac{105}{16}\cdot 2\left[\frac{1}{9}x^9 - \frac{2}{7}x^7 + \frac{1}{5}x^5\right|_0^1 = \frac{1}{3}\mathrm{nm}^2.$$

The expectation value of p^2 is performed using the relation $p^2 = -\hbar^2\partial^2/\partial x^2$, yielding

$$\langle p^2\rangle = -\frac{105}{16}\hbar^2\int_{-1}^{1} \mathrm{d}x(x^3 - x)\,\frac{\partial^2}{\partial x^2}(x^3 - x) = -\frac{105}{16}\hbar^2 \cdot 2\int_0^1 \mathrm{d}x\,(x^3 - x)6x = -\frac{105}{16}\hbar^2\cdot 2\cdot 6\left[\frac{1}{5}x^5 - \frac{1}{3}x^3\right|_0^1 = \frac{21}{2}\hbar^2.$$

(d) The uncertainties for the position and momentum operators are

$$\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \frac{1}{\sqrt{3}}\mathrm{nm},$$

and

$$\Delta p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2} = \sqrt{\frac{21}{2}}\hbar.$$

The Heisenberg uncertainty principle is satisfied

$$\Delta x \Delta p = \sqrt{\frac{7}{2}}\hbar \geq \frac{1}{2}\hbar.$$

(e) Inside the well, the potential is zero, so that $E = K = p^2/2m_\mathrm{e}$, and as a consequence

$$[H, p^2] = 0.$$

Hence, $\langle p^2 \rangle$ is independent of time (integral of motion) and

$$E = \frac{1}{2m_\mathrm{e}}\frac{21}{2}\hbar^2 \approx 5.25\frac{\hbar^2}{m_\mathrm{e}}.$$

Note that $\psi(x)$ can be considered as a *trial* wavefunction approximating the first excited state wavefunction ($n = 2$) for a box of width $L = 2$, as it can be verified by expanding the wavefunction in a Taylor series around $x = 0$

$$\phi_2(x) = \sin(\pi x) \approx \pi x - \frac{1}{3!}\pi^3 x^3$$

corresponding to energy

$$E_2 = \frac{1}{2m_\mathrm{e}}\pi^2\hbar^2 \approx 4.935\frac{\hbar^2}{m_\mathrm{e}}.$$

Indeed, the *trial* wavefunction just slightly overestimates the exact energy of the first excited state.

4. The $2s$ state of Hydrogen (3 + 5 + 3 = 11 points)

The radial part of the wavefunction of the $2s$ state of the Hydrogen atom (with quantum numbers $n = 2$ and $\ell = 0$) reads

$$R_{20}(r) = \frac{1}{2\sqrt{2}a_0^{3/2}} \left(2 - \frac{r}{a_0}\right) e^{-\frac{r}{2a_0}}.$$

Unlike the ground state wavefunction, this state displays a *forbidden* radius (node) r_0 for which the probability density to find the electron at a distance r_0 from the nucleus is zero.

(a) Find the position of the node r_0. (3pt)

(b) Find the most probable value $\bar{r}$ that maximizes the radial probability density. (5pt)
Hint: Maximizing the natural logarithm of the radial probability density is simpler than brute-force maximizing the function itself. Define $x = r/a_0$, and use

$$\frac{d}{dx}\log\left(f(x)\right) = \frac{f'(x)}{f(x)} = 0 \qquad \Longrightarrow \qquad f'(x) = \frac{d}{dx}f(x) = 0.$$

This procedure yields solutions $r_\pm$ (two local maxima, one of which is the global maximum) while the solutions ($r = 0$, and the node $r = r_0$) corresponding to minima are excluded.

(c) Sketch the radial probability density function, aiming at getting the overall shape and the relative positions of the node and the maxima correctly. (3pt)

Give the answers for r_0 and $\bar{r}$ parametrically in terms of the Bohr radius a_0.

Hint: The volume integral in spherical coordinates is given by:

$$\int dV = \int_0^\infty \mathrm{d}r\, r^2 \int_0^\pi \mathrm{d}\theta \sin(\theta) \int_0^{2\pi} \mathrm{d}\phi.$$

Solution

The radial probability density for this orbital is given by

$$P_{20}(r) = r^2 |R_{20}(r)|^2 = \frac{1}{8a_0^3} r^2 \left(2 - \frac{r}{a_0}\right)^2 e^{-\frac{r}{a_0}},$$

where the factor r^2 comes from the Jacobian of the transformation to polar coordinates.

The node condition $P_{20}(r) = 0$ yields a the (trivial) solution $r = 0$ and the actual node $r_0 = 2a_0$.

In order to find $\bar{r}$ we get the condition (apart from an irrelevant constant)

$$0 = \frac{d}{dr}P_{20}(r) \propto f(r) = r^2 \left(2 - \frac{r}{a_0}\right)^2 e^{-\frac{r}{a_0}}.$$

Using the suggested trick, we define $x = r/a_0$ and

$$\log\left(f(x)\right) = 2\log(x) + 2\log(2 - x) - x$$

differentiating we find

$$0 = \frac{d}{dx}\log\left(f(x)\right) = \frac{f'(x)}{f(x)} = \frac{2}{x} - \frac{2}{2 - x} - 1$$

and multiply by $x(2 - x)$ both sides, with the conditions $x \neq 0$ (origin) and $x \neq 2$ (node) yields

$$x^2 - 6x + 4 = 0$$

with solutions $x_\pm = 3 \pm \sqrt{5}$ or $r_\pm = (3 \pm \sqrt{5})a_0$, which correspond to a *local* and a *global* maxima.

A plot of the radial distribution function is provided below. It can be evaluated that the probability of finding the electron before or after the node are

$$\mathcal{P}(0 \leq r \leq 2a_0) \approx 0.053,$$

and

$$\mathcal{P}(r \geq 2a_0) \approx 0.947,$$

which verify the condition (conservation of probability)

$$\mathcal{P}(0 < r \leq r_0) + \mathcal{P}(r \geq a_0) = 1.$$

This can be observed in the very uneven distribution of the probability density.

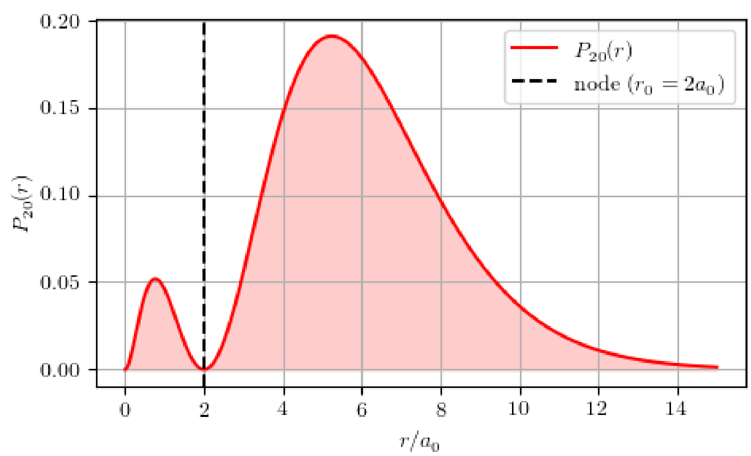

Figure 1: Radial probability density $P_{20}(r)$ for the Hydrogen 2s orbital as a function of the adimensional scale r/a_0. The dashed black line indicates the radial node at $r_0 = 2a_0$. The maxima are found for $r_\pm = (3\pm\sqrt{5})a_0$, corresponding to $r_- \approx 0.76a_0$ and $r_+ \approx 5.24a_0$.